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RESEARCH ARTICLE

Leveraging camera traps and artificial intelligence to explore thermoregulation behaviour

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Abstract

1. Behavioural thermoregulation has critical ecological and physiological consequences that profoundly influence individual fitness and species distributions, particularly in the context of climate change. However, field monitoring of this behaviour remains labour-intensive and time-consuming. With the rise of camera-based surveys and artificial intelligence (AI) approaches in computer vision, we should try to build better tools for characterizing animals' behavioural thermoregulation.
2. In this study, we developed a deep learning framework to automate the detection and classification of thermoregulation behaviour. We used lizards, the Rough-tail rock agama (*Laudakia vulgaris*), as a model animal for thermoregulation. We colour-marked the lizards and curated a diverse dataset of images captured by trail cameras under semi-natural conditions. Subsequently, we trained an object-detection model to identify lizards and image classification models to determine their microclimate usage (activity in sun or shade), which may indicate thermoregulation preferences. We then evaluated the performance of each model and analysed how the classification of thermoregulating lizards performed under different solar conditions (sun or shade), times of day and marking colours.
3. Our framework's models achieved high scores in several performance metrics. The behavioural thermoregulation classification model performed significantly better on sun-basking lizards, achieving the highest classification accuracy with white-marked lizards. Moreover, the hours of activity and the microclimate choices (sun vs shade-seeking behaviour) of lizards, generated by our framework, are closely aligned with manually annotated data.
4. Our study underscores the potential of AI in effectively tracking behavioural thermoregulation, offering a promising new direction for camera trap studies. This approach can potentially reduce the labour and time associated with ecological data collection and analysis and help gain a deeper understanding of species' thermal preferences and risks of climate change on species behaviour.

KEYWORDS
artificial intelligence, camera traps, climate, deep learning, lizard, shade

1 | INTRODUCTION

Climate conditions are among the most influential factors determining the survival and prosperity of animals within their natural environments. Different environments present various climatic challenges to animals, with extreme cold or heat leading to thermal stress, injury or even death (Angilletta, 2009; FalcónBrindis et al., 2020; Sunday et al., 2014). To minimize thermal risks and optimize performance, animals have developed a myriad of anatomical, physiological and behavioural adaptations (Angilletta, 2009). Often, behavioural thermoregulation is the most efficient strategy, wherein animals behaviourally aim to maintain their body temperature (in ectotherms; Bogert, 1949) or minimize the costs of heat production (in endotherms; Rezende & Bacigalupe, 2015). These strategies shape species' activity patterns, predator–prey interactions and competition worldwide (Angilletta, 2009). Understanding these behavioural strategies is becoming increasingly critical as our climate changes. Some animals may adapt by seeking shade, becoming more nocturnal or employing specific physiological mechanisms such as water immersion, panting and sweating (Dunkin et al., 2013; Kearney et al., 2009; Levy et al., 2019; Mitchell et al., 2018), others may still suffer from population declines (Levy et al., 2016; Sinervo et al., 2010; Stark et al., 2023). A better understanding of the thermal constraints facing animals is necessary for projecting animals' responses, identifying sensitive species and planning mitigation efforts. However, accurately observing and quantifying these strategies presents significant challenges.

While observations of behavioural thermoregulation are key to interpreting how animals respond to varying conditions, technological advancements for tracking and measuring these behaviours lag behind. Existing methods like manual observations (Huey & Pianka, 1977), temperature-sensitive loggers (Moore et al., 2018) and radiotelemetry (Ørskov et al., 2019) often prove logistically complicated, require capturing animals or are unsuitable for small species. Consequently, our capacity to explore the complexity of behavioural thermoregulation, the conditions influencing these behaviours and the potential impacts of ecological interactions, habitat loss and climate change is limited. In response to these challenges, camera traps have emerged as a promising tool to fill the observational gap left by traditional methods.

Camera traps have become an indispensable tool in wildlife research (O'Connell et al., 2011). They have been used to monitor animal populations (Silveira et al., 2003), track movements (Kays et al., 2009) and study habitat use (Alexy et al., 2003; Bowkett et al., 2008). The rapid expansion in camera-traps projects (Blount et al., 2021) leads to massive volumes of data, facilitating the development of artificial intelligence (AI) models for efficient data processing and analysis. Artificial intelligence-driven image recognition algorithms can now automatically detect animals (Beery et al., 2019), classify specific species (Norouzzadeh et al., 2021) and identify behaviours (Norouzzadeh et al., 2018). However, the potential of camera traps to study behavioural thermoregulation, which human observers can typically discern, remains underexplored (Norouzzadeh et al., 2018). Tapping into this potential through AI-based thermoregulation recognition could dramatically enhance our ability to investigate thermoregulation in the wild via large-scale, long-term observations.

In this study, we demonstrate how camera traps can be employed to monitor the behavioural thermoregulation of animals. Specifically, we developed a modelling framework for training an animal detection model and an animal classification model, designed to recognize whether an animal is active in sunny or shaded microhabitats. Using a case study of lizards in semi-natural conditions, we illustrate the construction and testing of this framework and discuss the challenges that impact model performance. This framework can be easily adapted to other scenarios, including natural conditions and varying animal sizes. We believe the integration of AI for identifying thermoregulation behaviours will add a vital climatic dimension to camera-based studies and enable the analysis of previously published image datasets.

2 | MATERIALS AND METHODS

2.1 | General approach

The AI framework employed in this study aims to automatically identify animals' thermoregulation behaviour by first detecting their location in each image and then classifying their behaviour (see Figure 1). For simplicity, we focus on determining whether an animal is situated in a sunny or shaded spot, as this behaviour is very common to terrestrial animals, including both ectotherms and endotherms. We began by training and validating the framework using thermoregulating lizards as a model system, subsequently assessing the model's performance across different marking colours and climate conditions. Marking colours are employed in many studies for individual identification (e.g. Brown, 1997). However, they may also resemble the skin, fur or plumage colours of other species, possibly affecting the performance of the thermoregulation model. To achieve these goals, we followed a three-step process: First, we collected empirical images of active animals under semi-natural conditions; second, we trained the detection and classification models and finally, we evaluated the models' performance in various microhabitats (sun or shade) and at different times of day (morning, noon and afternoon).

2.2 | Study species

Our model species was the Rough-tail rock agama (*Laudakia vulgaris*), previously known as *Stellagama stellio*. The species is a moderate-sized lizard, with a maximum snout-vent length (SVL) of up to 210 mm and a mass of up to 150 g (Karameta et al., 2017). It inhabits diverse habitats ranging from Egypt, Jordan, Israel and southwestern and western Syria (Amr et al., 2012; Karameta et al., 2022). The species' diet includes arthropods, plants and occasionally smaller lizards, snakes and birds (Bar et al., 2021; Ibrahim & El-Naggar, 2013;

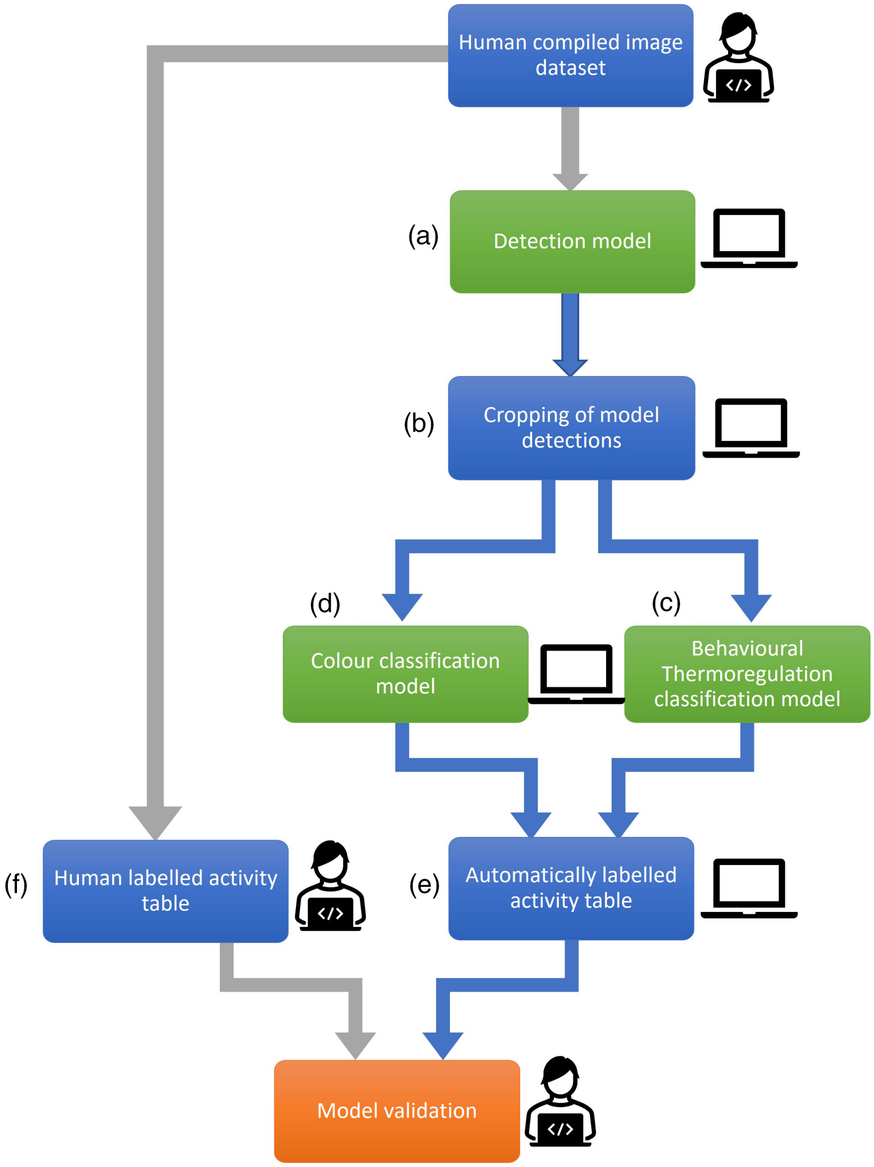

FIGURE 1 A schematic representation of the framework and its validation. During the first step, the workflow scans images taken by camera traps and identifies the animals (a). In the second stage, the detected animals are cropped from the images (b) and sent to thermoregulation-behavioural (c) and colour-identification (d) models. The results are compiled in an activity table for data analysis (e). For a test dataset, we created a manually labelled activity table (f) and compared it with the corresponding table from our framework. Green boxes represent models, and blue boxes represent intermediate steps. Blue arrows represent the framework, and grey arrows represent human labour.

Karameta et al., 2015). The species is diurnal and heliothermic, preferring a body temperature of 35.9–37.7°C (Saber, 2012). Field body temperatures have been observed to range from 31.6 to 38.6°C at low elevations and from 23.4 to 39.2°C at high elevations (Hertz et al., 1983; Hertz & Nevo, 1981), with recorded temperatures reaching up to 41.7°C in some instances (personal observation). *Laudakia vulgaris* can be observed basking throughout the day, even during the hottest hours (Bar et al., 2021) and is known to thermoregulate by shuttling between sun and shade microenvironments (Norfolk et al., 2010). These characteristics make *L. vulgaris* an excellent model species for studying behavioural thermoregulation through camera trap surveys.

2.3 | Lizard collection and treatment

Adult *L. vulgaris* were collected from two distinct populations in Northern Israel between October and December 2020. These two populations were also part of another observational study. Six individuals (2 males, 4 females) were captured from a forest near Yokneam (32°38′27.9″ N 35°05′54.6″ E), located 200 m above sea level with an average annual air temperature of 19.7 ± 0.1°C from 2015 to 2022. The remaining nine (3 males, 6 females) were captured on Hermon Mountain (33°17′22.2″ N 35°45′35.8″ E), 1500 m above sea level, where the mean annual air temperature was 12.2 ± 0.1°C from 2019 to 2022. Post-collection, the lizards were housed in the reptile section of Meier Segal's Garden for Zoological Research in Tel Aviv. We set up five semi-natural enclosures, each housing two females and one male. The enclosures measured 245 cm × 155 cm, enabling the lizards access to natural sunlight and a variety of microclimatic conditions throughout the day. The 50 cm high concrete walls of the enclosures provided both sunny and shaded areas. For shelter, bricks were placed at each corner of the enclosures, and a tile was positioned at the centre of each enclosure for basking.

Lizards were fed twice a week in the spring and fall, and every 1–2 weeks in winter, except for rainy days when they were generally inactive. The diet comprised primarily of large mealworms (*Tenebrio molitor* and *Zophobas morio*), with occasional supplements of locusts (*Schistocerca gregaria*), German cockroaches (*Blattella germanica*), vegetables and hibiscus flowers from Meier Segal's Garden. Water was made available ad libitum throughout the year. All animal procedures in this study were conducted with the approval of the Ethical Committee of Tel-Aviv University (04-21-060) and The Israel Nature and Parks Authority (2020/42542, 2021/42779).

2.4 | Image collection

To observe and document the lizards' thermoregulation behaviour, we installed two wide-angle camera traps (LTL ACORN Ltl-5310WA, AcornCamera, Shenzhen, China) on the northern and southern walls of each enclosure. Each camera has a 100-degree field of view and a 12-megapixel resolution and was set to capture three consecutive images upon detecting lizard movement, with no delay between images. For individual identification, we marked each lizard in every enclosure with a distinct nail polish colour: blue, red or white. While this colouring might affect skin reflectance and, consequently, behavioural thermoregulation (Pearson & Bradford, 1976), it also enabled us to investigate how colouration influences the performance of our models.

2.5 | A deep learning framework for detecting thermoregulation

The framework we employed consists of two stages (Figure 1). The first stage involves scanning a directory of images to identify any lizards present. If the framework detects a lizard in an image, the second stage initiates, classifying the observed microhabitat chosen by the lizard as either sunny or shaded. Once both stages are completed, the framework records the results for the image in a table for further analysis.

2.5.1 | Stage one: Lizard detection model

Model description

We employed an object detection model to identify lizards within the images. Object detection models are neural networks trained to detect and classify objects in images, and they have widespread applications such as in self-driving cars, surveillance systems and image search engines (Fergus et al., 2005; Himeur et al., 2023; Zaidi et al., 2022). Since existing models for animal detection, such as the MegaDetector (Beery et al., 2019; Microsoft, 2020), are not yet effective at detecting reptiles, we trained a pre-trained object detection model, Faster R-CNN (Ren et al., 2017), to recognize lizards. Given an image, the detection model identifies one or more objects of interest, providing their bounding boxes to the user.

Model training

In general, the training dataset for Faster R-CNN consists of a set of images and their corresponding annotations, which indicate the locations of objects of interest. Each annotation includes the coordinates of a bounding box surrounding the object and a label specifying its class (i.e. lizard; Girshick, 2015; Jiang & Learned-Miller, 2017; Ren et al., 2017). To optimize the model's performance under different scenarios, it is essential to create an annotation dataset capturing diverse lizard characteristics, such as differences in size, backgrounds and locations within images. This diversity enables the model to accurately recognize lizards across various conditions, enhancing its generalization capabilities and overall performance.

Given the importance of a diverse and comprehensive annotation dataset for training the Faster R-CNN model, and considering the labour-intensive nature of manual annotation, we adopted a semi-automated method that incorporated an initial, preliminary detection model to streamline the annotation process (Figure 2, grey area). First, we hand-annotated 167 images using LabelImg, a graphical image annotation tool (Tzutalin, 2015), outlining bounding boxes around each lizard. For each image, LabelImg produced an XML file containing the coordinates and tags for every bounding box. We selected images from all enclosures and cameras, featuring lizards in various locations, seasons and activity hours. We also added 211 background images (without lizards) to train the model to recognize empty enclosures.

In the second stage, we employed the preliminary model to automatically generate annotations for a dataset comprising 53,573 images (Figure 2, yellow area). This model produced 30,038 annotations, which we imported into LabelImg to verify their accuracy. Through this semi-automated method, we identified an additional 1358 accurate annotations to include in the final training dataset (Figure 2, yellow area), which contained a total of 1525 annotations. This approach significantly reduced the time and effort required for manual annotation while ensuring the high quality and diversity of the annotations in the training dataset.

For the training process, we split the dataset into 80% for training and 20% for validation. During training, we utilized a learning rate of 0.005 for the first four epochs and then 0.001 for an additional 15 epochs with a batch size of 2. We monitored the loss and accuracy of the validation set to avoid overfitting.

2.5.2 | Stage two: Classification of Lizards' Thermoregulation Behaviour and Marking Colours

Model description

We trained two classification models utilizing the ResNet deep learning architecture (He et al., 2016). We used a pre-trained version of the model, with 101 layers, which was trained on more than a million images from the ImageNet database (Deng et al., 2009). Our first classification model was trained to classify lizards based

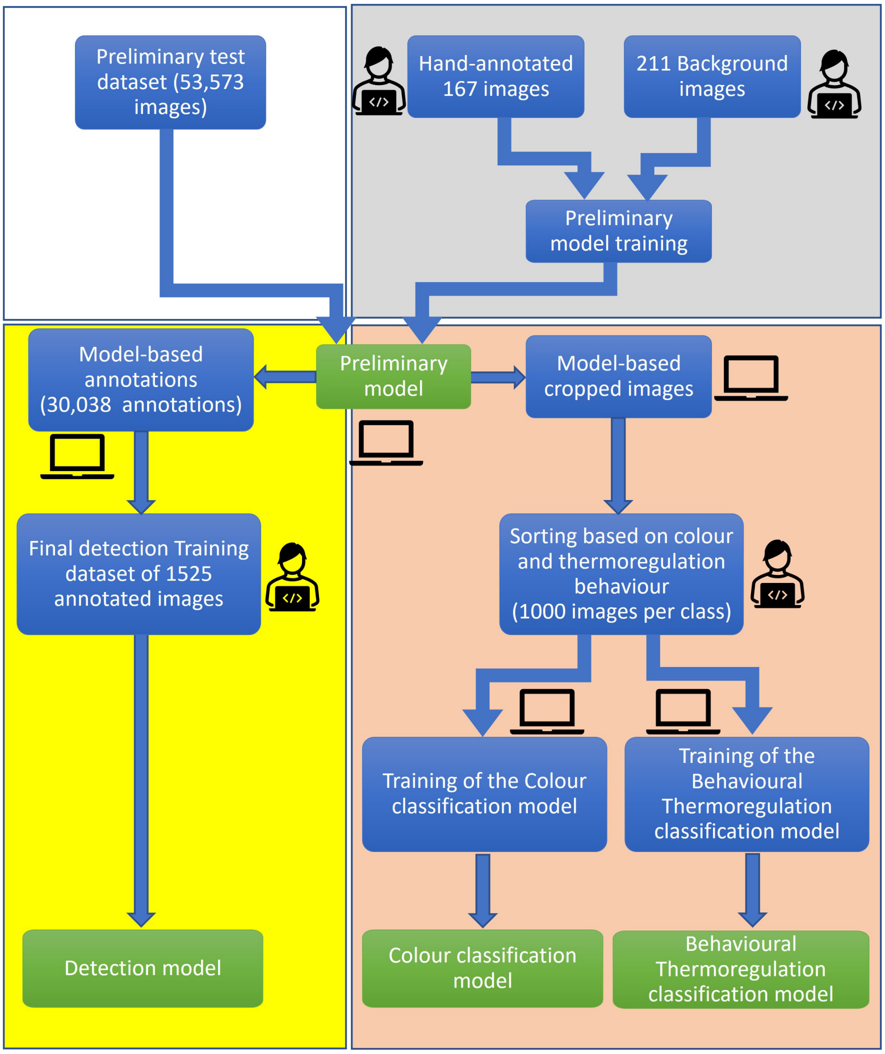

FIGURE 2 A schematic illustration of the training process for our object detection and classification models. Each stage of the process is represented by a separate area. Grey area—Training of a preliminary dataset for the detection model. White area—Creation of a large dataset to be sent to the preliminary detection model for automatic annotation. Yellow area—Use of the preliminary model to automate annotation generation for the larger dataset. Automated annotations are then manually verified and a final set of correct annotations is used to train the final detection model. Orange area—Use of the preliminary model to crop detected lizards. These automated crops are then sorted to create training datasets for the Thermoregulation and Colour classification models. Green boxes within each area indicate which models are used at each step, and blue boxes represent the intermediate steps. This entire process is designed to streamline the creation of a diverse and accurate training dataset for both the detection model and the classification models.

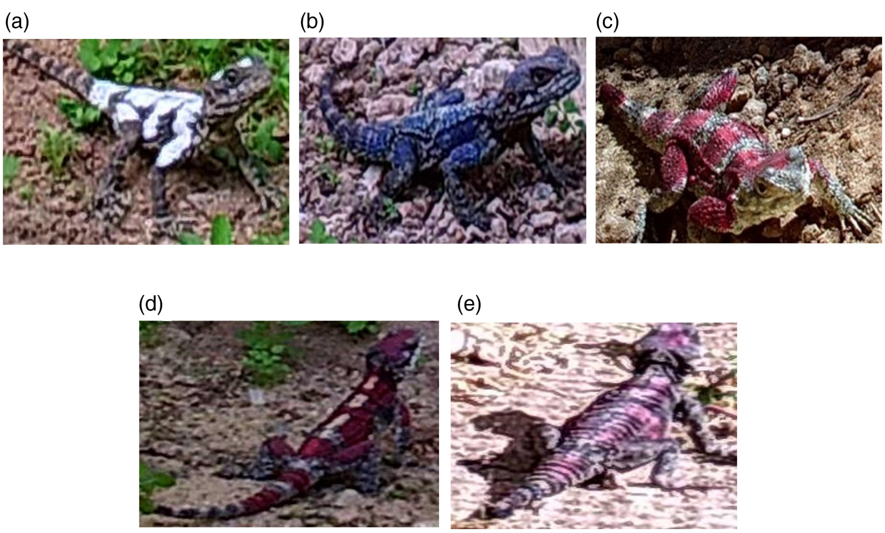

FIGURE 3 Examples of cropped images of detected lizards, showing different marking colours (a—white, b—blue, c—red), and shade conditions (d—shade, e—sun).

on their thermoregulation behaviour, specifically their microhabitat association—whether they were located in sunlit or shaded areas. The second model was trained to classify the colours of the lizards' markings.

Model training

Both classification models use cropped images as input, which we generated using the preliminary detection model (Figure 2, orange area). For the thermoregulation behaviour model's training set, we

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categorized the cropped images into two classes: (1) lizards in sunlit areas and (2) lizards in shaded areas (see examples in Figure 3). Each class included 1000 images for training. For the marking colour classification model, we sorted the cropped images into three classes, each representing a different marking colour. Each class contained 1000 images for training.

In addition, we integrated two helper classifications into our models. Specifically, to further minimize false positives overlooked by the detection model, we introduced a 'negative' class to the Behavioural Thermoregulation model. We trained this 'negative' class using 266 incorrect detections, a relatively small dataset due to the detection model's high accuracy. Also, to handle images where the lizard's marked colour was ambiguous, we added an 'unclear' class to the Colour classification model, using 109 such images for the training data. Since these helper classes were not part of the study's primary scope, we did not analyse their performance scores nor included them in the models' overall performance metrics.

For the training process of each model, we split the dataset into 80% for training and 20% for validation. During training, we utilized a learning rate of 0.001 for the first four epochs and then 0.0001 for an additional 20 epochs with a batch size of 16. We monitored the loss and accuracy of the validation set to avoid overfitting.

Both the detection and classification models were trained and run on a server equipped with 2 Nvidia Titan RTX Graphical Processing Units (GPU). Code was written in Python, utilizing the fastai (Howard & Gugger, 2020) and PyTorch (Paszke et al., 2019) libraries. Training followed the code and guidelines detailed in Microsoft's repository of computer-vision recipes (github.com/microsoft/computervision-recipes).

To examine which sections of the images help the classification models distinguish between the different classes, we employed Gradient Class Activation Mapping (Grad-CAM; Selvaraju et al., 2020). This method allows us to identify the most influential segments of an image for the model's prediction. We applied this approach to every combination of Thermoregulation (sun/shade) and Colour (white/blue/red) classes.

2.6 | Validation and performance evaluation

To evaluate our framework's performance, we compiled a validation table. This table contained our framework's predicted classifications for lizard characteristics (colour and sun/shade status) and the actual values, which were determined through manual labelling of the images. This allowed us to calculate the framework's accuracy, precision, recall and F1 scores (see below for scores' explanation), giving us an overview of how well it can identify behavioural thermoregulation in camera trap images (Figure 1).

2.6.1 | Preparing the human-labelled dataset

For model validation, we manually labelled the thermoregulation behaviour and colour of lizards during a sunny day in spring, a season of high activity for this species. The activity for the chosen day generated 5520 images in our enclosures. Each image was reviewed, and every time we identified an active lizard, we noted the shade or sun selection, marking colour, time of day, enclosure, camera name, image filename and any additional remarks. If multiple lizards were observed in the same image, each was given a separate row in the table. The final human-labelled table contained 6835 rows. To assess the performance of our detection and Behavioural Thermoregulation classification models on unmarked lizards, which more closely mimic the conditions researchers encounter in natural habitats, we compiled data from 203 images into a validation table. These images, containing a total of 840 lizards, were captured by our camera traps in the enclosures before the lizards were marked (January 2021).

Additionally, we manually estimated the hourly percentage of shade availability in each enclosure using our camera-trap images from each enclosure. While this analysis falls outside the study's primary scope, it provides valuable context for comparing lizards' microhabitat selection. Specifically, it helps to determine whether lizards actively thermoregulate or if they simply occupy the most readily available sunlit or shaded areas (Huey & Pianka, 1977).

2.6.2 | Prediction generation and comparison

We applied the trained framework to the images used in the human-labelled table. For each image, where a lizard was detected, the framework generated a text file containing the predicted classifications (marking colour and shade/sun). These text files were processed to create a table with the predicted classifications, a confidence score for each classification, time of day, enclosure name, camera name, image filename and coordinates of the corresponding bounding box for each detected lizard. To construct the final validation table, we combined the predictions from our framework with the corresponding human-labelled observations for each detected lizard.

2.6.3 | Prediction performance analysis

Several metrics can be used to evaluate an AI model's performance, providing a comprehensive understanding of how well the model works. Accuracy (Equation 1) is a commonly used metric that measures the proportion of correct predictions made by the model out of the total number of predictions (Dinga et al., 2019).

$$\text{Accuracy} = \frac{\text{true positives} + \text{true negatives}}{\text{true positives} + \text{false positives} + \text{true negatives} + \text{false negatives}} \quad (1)$$

While accuracy is a useful metric for overall model performance, it may not be the best choice for models with imbalanced classes–where one class size significantly outweighs another (Juba & Le, 2019). For instance, in our Thermoregulation Behaviour model, we may have fewer images of lizards in the shade than in the sun, creating an imbalance that may bias the accuracy metric. In such scenarios, precision, recall and the F1 score are better-suited evaluation metrics.

Precision measures the proportion of true positives out of all positive predictions (Equation 2). A high precision score indicates that the model is accurately identifying true positives while limiting false positives (e.g. mistakenly identifying a non-lizard object as a lizard). On the other hand, recall quantifies the proportion of true positives out of all actual positive instances in the dataset (Equation 3). A high recall score suggests that the model is capturing most of the actual positive instances and minimizing false negatives (e.g. missing a lizard in the image) (Ditria et al., 2020; Singh, 2022).

$$\text{Precision} = \frac{\text{true positives}}{\text{true positives} + \text{false positives}} \tag{2}$$

$$\text{Recall} = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}} \tag{3}$$

The F1 score is a weighted harmonic mean of precision and recall (Equation 4; Dinga et al., 2019; Zhang et al., 2019). It effectively balances precision and recall and is especially useful when classes are imbalanced. It considers both false positives and false negatives, with a high score indicating accurate identification of true positives.

$$\text{F1 score} = \frac{2 \times (\text{Precision} \times \text{Recall})}{(\text{Precision} + \text{Recall})} \tag{4}$$

To evaluate our framework, we computed precision, recall and F1 scores for each model, utilizing our validation tables for both marked and unmarked lizards. Additionally, we calculated confusion matrices to detail each model's true positives, true negatives, false positives and false negatives.

2.7 | Statistical investigation of model performance

Using our validation table for marked lizards, we examined variations in the performance of the Thermoregulation Behaviour model through a Bayesian hierarchical logistic regression analysis. We considered the model's performance as a binary response variable (0—correct prediction, 1—incorrect prediction), and the true lizard location (sun or shade), time of day (morning—before 11:00, noon—between 11:00 and 15:00, afternoon—after 15:00) and marking-colour (blue, white or red) as categorical factors, including all possible interactions between them.

To account for repeated measurements taken from the same cameras and enclosures, we included camera and enclosure names as a nested random effect in the model. For each camera, we also added an AR1 autocorrelation term to the model to account for temporal autocorrelation between images captured at similar times. Both fixed and random effects used non-informative priors. We fitted the model using the PyMC python package (Salvatier et al., 2016), ran it using an No-U-Turn (NUTS) sampler (Homan & Gelman, 2014) with 1000 iterations and three chains and assessed convergence using trace plots, the Gelman-Rubin $\hat{R}$ diagnostic and the effective sample size.

To evaluate the significance of the predictors, we examined the posterior distributions of the coefficients and calculated their 95% credible intervals. A coefficient was deemed significantly different from zero at the 5% level if its credible interval did not overlap with zero.

2.8 | Comparison of framework-labelled and human-labelled activity patterns

Using our validation table of marked lizards, we quantified the hourly proportions of time lizards were observed in sunny and shaded microhabitats. We first segregated the data into distinct events, defined as instances where the same lizard was captured within 10 s in the same microhabitat (sun or shade). We then calculated the duration of each event by subtracting its start and end times. Next, for each enclosure and hour, we calculated the proportions of time lizards were captured active in each microhabitat. We used this method to generate two activity pattern tables, classifying the lizards' microhabitats either by our framework or by our manual labelling.

Since our cameras were motion-activated, they did not capture periods when lizards remained stationary, such as during basking. This limitation prevented us from determining the total time lizards spent in each microhabitat or tracking their transitions between microhabitats (i.e. assuming lizards stayed in the last microhabitat captured by the camera, Stark et al., 2024). Despite these challenges, our analysis enabled a valuable comparison between human and AI labelling of animal movements within sun-exposed or shaded microhabitats.

3 | RESULTS

3.1 | Lizard detection

We applied the detection model with a confidence score threshold of 50%, resulting in 6483 detections. Of these, 6223 were correct detections of lizards, whereas 260 were false positives (wrongly identified as lizards), and 660 were false negatives (not detected lizards). The model demonstrated an accuracy of 87.12%, a precision of 95.99%, a recall of 90.41% and an F1 score of 93.12%. For unmarked lizards, the model demonstrated an accuracy of 91.39%, a precision of 92.77%, a recall of 98.40% and an F1 score of 95.50%.

3.2 | Classification of Lizards' Behavioural Thermoregulation and Marking Colours

The reliability of predictions from the Behavioural Thermoregulation and Colour classification models improved significantly with higher confidence scores (Figure 4). At confidence levels above 95% (dashed lines in Figure 4), the Thermoregulation model achieved 96.52%, 90.55%, 92.38% and 91.46% for accuracy, precision, recall and F1 score, respectively. The Colour classification model performed similarly well with 96.54%, 96.54%, 97.56% and 96.98% for accuracy, precision, recall and F1 score. Classifications with 95% confidence level or above encompass 85% of the Behavioural Thermoregulation and 80% of the Colour classification model's predictions. This combination of strong performance at confidence levels above 95% and the high proportion of images meeting this threshold suggests that setting a high confidence threshold for tracking thermoregulation behaviour allows us to retain most of the data while enhancing prediction reliability. Our Gradient Class Activation Mapping analysis revealed that while the Colour classification model focuses on the lizard itself, the Behavioural Thermoregulation model primarily attends to the proximate environment of the lizard (Figure 5a–c). Intriguingly, in images capturing both sunlit and shaded microhabitats, the Behavioural Thermoregulation model zeroes in on regions adjacent to the lizard's abdomen at the bottom of the image (Figure 5d).

The high performance of the models is also evident when comparing the framework- and human-labelled activity patterns. Specifically, our framework generated a table of activity patterns that closely resembled the manually compiled table (Figure 6).

The Behavioural Thermoregulation classification model performed well for unmarked lizards as well, achieving 97.96%, 97.67%, 96.55% and 97.11% for accuracy, precision, recall and F1 score, respectively (Figure 7).

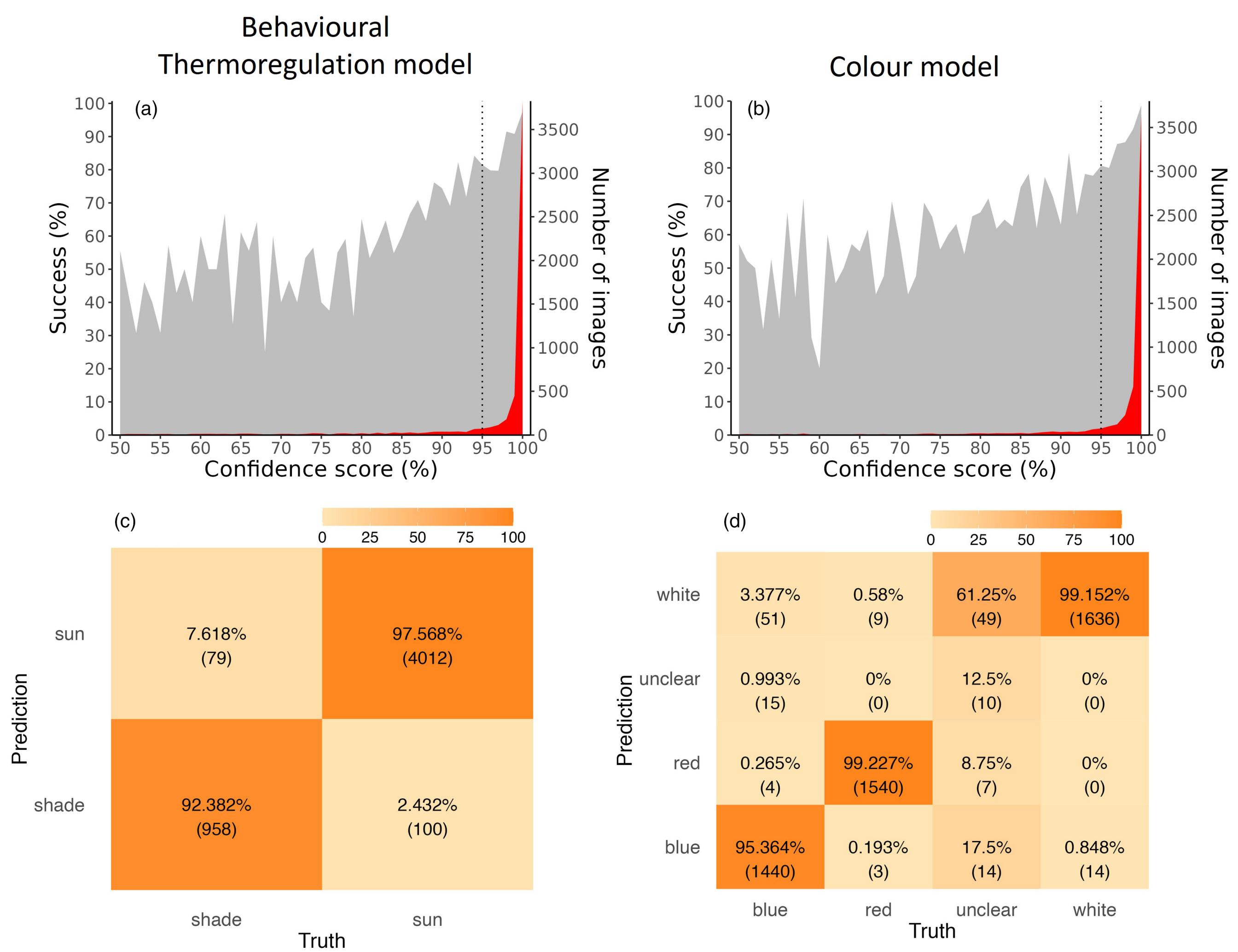

FIGURE 4 The performance of the classification models. The success percentages of the (a) Behavioural Thermoregulation and (b) Colour models increased with models' confidence scores (grey area), and model predictions for our images (red area) were almost exclusively above 95% confidence (dashed vertical line), where model performance is high. Confusion matrices of the models (c—Behavioural Thermoregulation, d—Colour) show true observations versus model predictions. In each cell of the confusion matrices, we present the proportion of cases within the same column, along with the number of cases, given in parentheses.

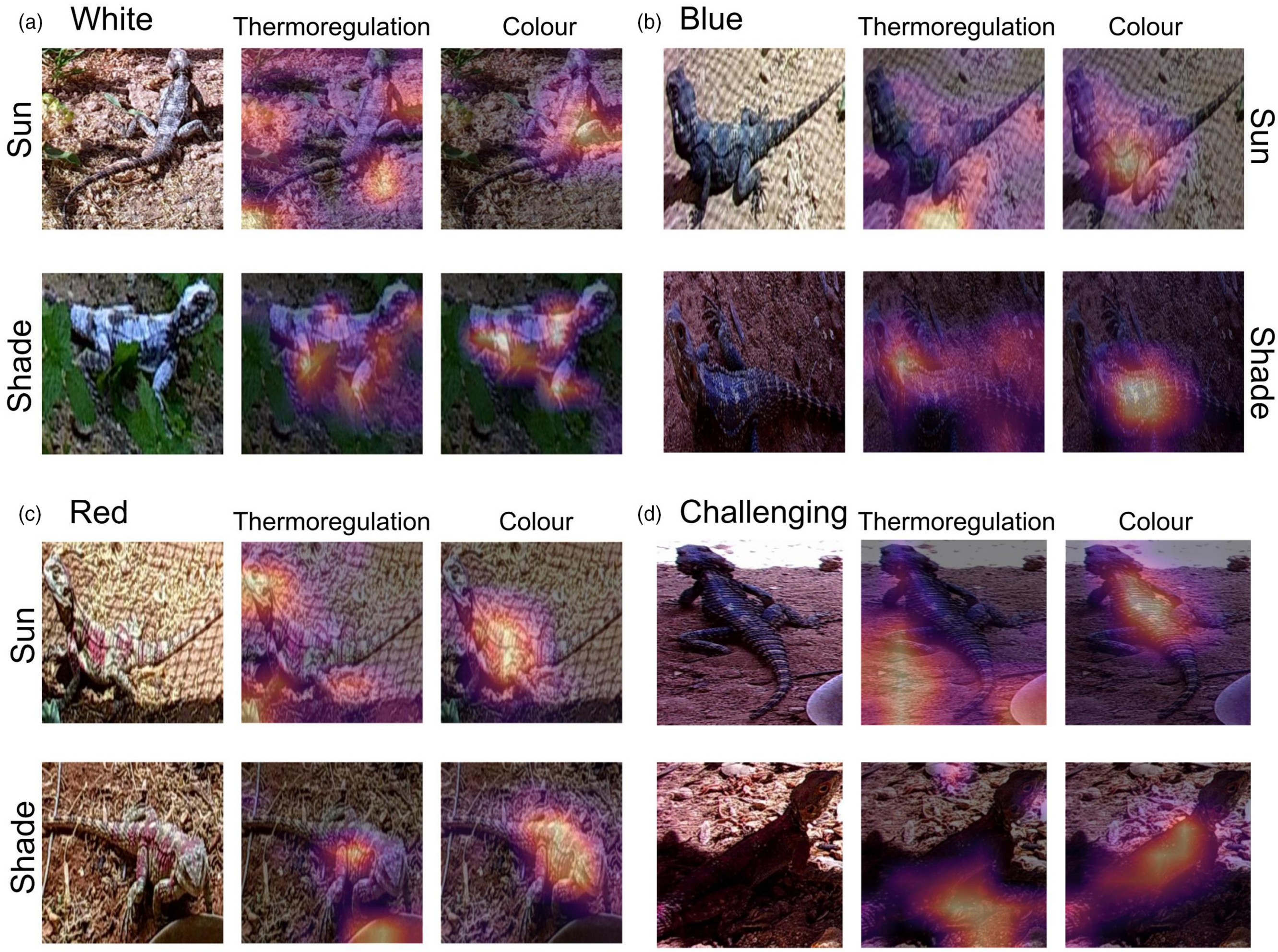

FIGURE 5 The Behavioural Thermoregulation model primarily discerns between sunlit and shaded microhabitats by focusing on the environmental context surrounding the animal. In contrast, the Colour classification model exclusively scrutinizes the lizard to ascertain its colour. This figure illustrates examples of white (a), blue (b) and red (c) lizards in sunlit (top row) and shaded (bottom row) conditions. Additionally, it presents complex cases (d) where the lizard is in the shade, but a portion of the image captures sunlit ground. Each panel (a–d) exhibits an original image (left), followed by heatmaps of the regions crucial for Behavioural Thermoregulation (middle) and Marking-Colour (right) classifications. The heatmaps were generated using Gradient Class Activation Mapping (Selvaraju et al., 2020), signifying the areas most influential for each classification task.

3.2.1 | Factors influencing the Behavioural Thermoregulation classification model

Our statistical model's estimates (Table 1) suggest that the performance of the Behavioural Thermoregulation model varies depending on both the environmental context and marking colour. Specifically, the model shows increased reliability when lizards are in the sun and generally performs better with white-marked lizards, except during the noon hours. Interestingly, the red marking colour impacted model performance in a more complex way: during noon, it enhanced the model's reliability in the shade but diminished it in sunlit conditions (Figure 8). At one particular camera, the framework incorrectly classified several lizards as active in the sun when, in fact, they were active in the shade. This discrepancy is intriguing as the activity was captured by one camera, and the lizards appeared to be in a sunlit area. However, simultaneous images taken by the second camera of the enclosure, which did not directly capture the lizards, clearly show that the areas where the lizards were observed were indeed shaded (Figure 9a,b).

4 | DISCUSSION

4.1 | The promise of camera traps and artificial intelligence for collecting thermoregulation data

Thermoregulation is a fundamental aspect of animal behaviour, playing a key role in their ability to thrive within diverse climatic conditions (Huey, 1982). Tracking how animals thermoregulate under different conditions is a challenging task, but AI algorithms show a promising ability to lead current approaches of many human and animal tracking technologies. Here, we successfully utilized these

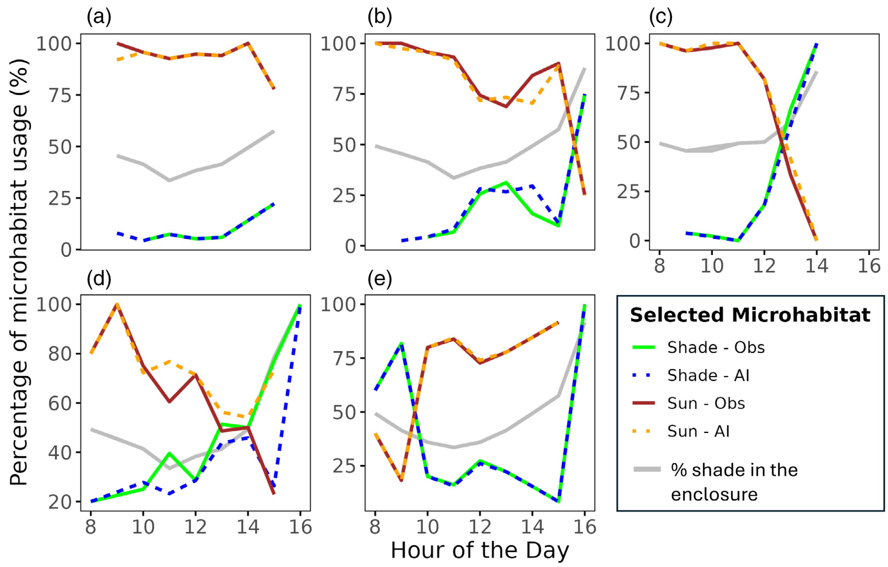

FIGURE 6 Our framework—and human-labelled images produced similar activity patterns in sunny and shaded microhabitats. We show the proportion of time that lizards spent in sunny and shaded microhabitats, as was found by our framework (dotted lines: orange—sun, blue—shade) and by a manual examination (solid lines: brown—sun, green—shade). Each panel (a-e) represent a different enclosure. Grey lines represent the proportion of shaded ground in each enclosure.

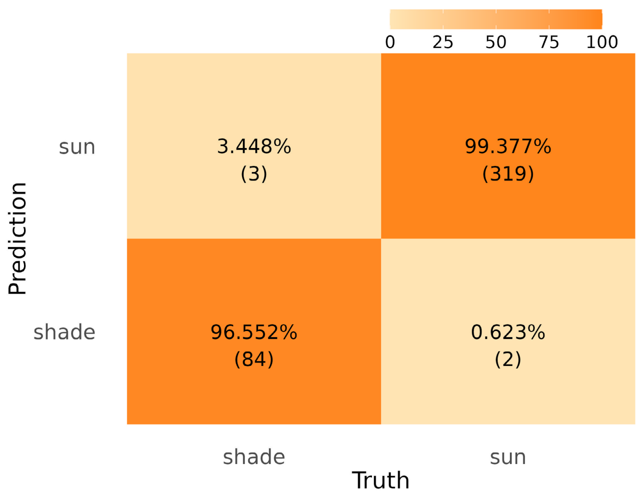

FIGURE 7 The performance of the Behavioural Thermoregulation classification model for unmarked lizards. The confusion matrix shows true observations versus model predictions. In each cell of the confusion matrices, we present the proportion of cases within the same column, along with the number of cases given in parentheses.

TABLE 1 Summary of the Bayesian model estimates with the model failures as the binary response variable.

	Estimate ± SD	2.50%	97.50%
Intercept	**−2.10 ± 0.91**	**−3.73**	**−0.31**
Morning	−1.16 ± 0.83	−2.85	0.30
Noon	−0.61 ± 0.56	−1.66	0.46
Red	−0.15 ± 0.46	−0.99	0.72
White	**−3.85 ± 0.77**	**−5.33**	**−2.45**
Sun	**−2.27 ± 0.59**	**−3.29**	**−1.09**
Morning × red	−0.28 ± 1.00	−2.15	1.59
Morning × white	−0.98 ± 1.64	−3.93	2.25
Noon × red	**−1.77 ± 0.68**	**−3.01**	**−0.47**
Noon × white	**3.04 ± 0.82**	**1.56**	**4.58**
Morning × sun	1.34 ± 0.92	−0.51	3.00
Noon × sun	−1.18 ± 0.74	−2.55	0.18
White × sun	−0.55 ± 0.88	−2.16	1.14
Red × sun	0.20 ± 1.05	−1.67	2.23
Morning × red × sun	−0.24 ± 1.24	−2.53	2.04
Morning × white × sun	−0.89 ± 1.76	−4.26	2.35
Noon × red × sun	**4.71 ± 1.03**	**2.73**	**6.59**
Noon × white × sun	0.28 ± 1.15	−1.99	2.32

Note: We show the model posterior distribution of the categorical predictors as estimated coefficients (estimate ± SD) and 95% credible intervals. Significant terms, where zero is outside of 95% credible intervals, are in bold.

algorithms to develop better techniques for tracking a critical element of animals' behaviour—shuttling between sun-exposed and shaded microhabitats. We show that off-the-shelf classification models can be trained to accurately classify whether an animal is in the sun or shade with a relatively small training dataset. As cameras are extensively used in ecological research, our framework demonstrates that AI algorithms can not only track animal behaviour, such as activity patterns, but also integrate the selection of microclimates (Figure 6), adding a physiological context to such studies.

Camera traps have revolutionized wildlife tracking (Fennell et al., 2022; O'Connell et al., 2011; Vélez et al., 2023), amassing an unprecedented volume of data for researchers. This abundance of data has precipitated the evolution of various artificial intelligence (AI) models and frameworks tailored for ecological analysis (reviewed by Fennell et al., 2022). Past models have proficiently classified animal species (Norouzzadeh et al., 2018) and, more recently, identified individual animals, from giant pandas (*Ailuropoda melanoleuca*) (Chen et al., 2020), eight species of bears (Clapham et al., 2022), to 14 species of primates (Deb et al., 2018). However, these models primarily concentrate on the subject animal, often overlooking the encompassing environmental context. Our

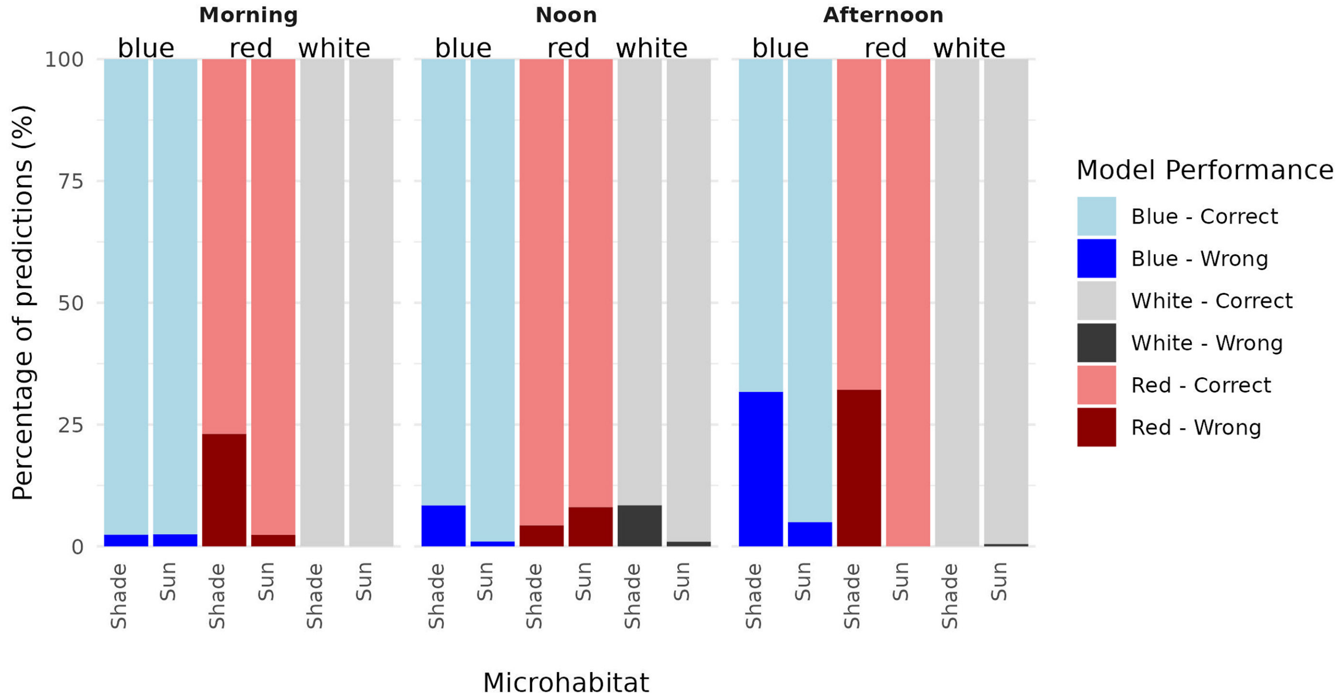

FIGURE 8 The performance of the Behavioural Thermoregulation classification model varies with time of day, colour and microhabitat. The chart shows the percentage of correct (light-coloured) and incorrect (dark-coloured) classifications.

FIGURE 9 Four examples of incorrect predictions made by the framework. In each image, we marked the misclassified lizard with a red box and enlarged this part of the image in the right-bottom corner of the image. In each red box, the green box represents the bounding box of the lizard, as found by the detection model. In (a), a lizard appears to be in a sunny microhabitat, but the elevated rock is actually in the shade, as captured by the second camera in the enclosure (b). In (c), the lizard is in a sunny microhabitat, but the enclosure's top net casts shade on the lizard, causing the model to misclassify the lizard as if it were in the shade (red box). In (d), only the tail of the lizard is visible in the image, and the Colour classification model misclassifies the lizard as white-marked (red box). Additionally, the detection model failed to detect one of the lizards in the image (blue box).

framework, on the other hand, demonstrates that a classification model can discern between the animal and the environmental context surrounding it (Figure 5). This ability of the model constitutes a novel proof of concept, positing that the background composed of different microhabitats or the conjunction of the animal and its environment can impart essential ecological information. To our knowledge, this is the first application of such a concept for these models.

Amidst climate change and its escalating threats to natural populations, our framework can generate data from camera traps that can help identify and mitigate these risks (Buchholz et al., 2021). Specifically, these thermoregulation data can be harnessed to decipher when and where animals appear and to untangle the conditions that drive species to seek the sun's warmth or prefer the coolness of the shade. This knowledge could illuminate preferred environmental conditions, underscore the significance of shading elements and contribute to other essential ecological insights. For instance, Wolff et al. (2020) employed camera traps to track how the preference of deer (*Odocoileus virginianus*) for shaded or unshaded feeders fluctuated with climatic conditions. Similarly, camera trap data can reveal animals' thermal preferences for activity. For example, Cserkész et al. (2023) used camera traps to show that the endangered Hungarian birch mouse (*Sicista trizona*) can be active even at −6°, a much cooler condition than previously thought. Moreover, using camera trap videos, Stark et al. (2024) highlighted the importance of thermoregulation and movement costs in desert lizards. Such insights into thermal preferences and tolerances are invaluable for parameterizing mechanistic niche models designed to predict animal survival under varying climatic conditions (Briscoe et al., 2023; Stark et al., 2023). This enhanced understanding of microhabitat preferences can inform the design of wildlife reserves, incorporating a diversity of microhabitats to support different species amidst climate change (Stark et al., 2023). Additionally, this knowledge can guide management and habitat restoration strategies, such as strategically placing shade structures or watering stations in habitats facing the challenges of rising temperatures.

4.2 | Criticial lessons learned from our framework

Our Thermoregulation and Colour classification models have demonstrated high accuracy, even with smaller training datasets compared to previous studies (Guo et al., 2020; Norouzzadeh et al., 2018; Willi et al., 2019). Moreover, the models exhibited a strong correlation between accuracy and confidence scores, with a particularly high success rate observed at 95% confidence and above. This suggests that pre-trained models, like ResNet (He et al., 2016), can significantly contribute to high accuracy, especially when working with smaller datasets. This is because these models have already been trained on large, diverse datasets (such as ImageNet, Deng et al., 2009), and have thus learned to recognize a rich set of features (Yosinski et al., 2014). However, some factors affected the model's performance, such as the lizard's marking colours, time of day and sunlight conditions (Table 1, Figure 8). For example, it is possible that the Thermoregulation model more accurately classifies lizards in the sunny microhabitat since the contrast between the lizard and the substrate is greater. Classification models might have been more successful for white-marked lizards since the contrast between them and the substrate is relatively higher than in other marking colours. On some occasions, the model mistakenly classified lizards as being in the shade, possibly due to shadows cast by the enclosure's iron net or the illusion created by the sun's angle (Figure 9c). These issues could be resolved by altering the height and position of enclosures and cameras, reducing bias from the sun's angle. We are not sure why the red marking was more challenging to the Thermoregulation model, especially in the shade (Figure 8), but it is possible that the resemblance of the red colour to the colour of the background soil has focused the model's attention on the lizard rather than on the ground surface (Figure 5c). Nevertheless, our findings suggest that to better understand the influence of different colours and species on the Thermoregulation model, further testing on a variety of colours and species is recommended.

Researchers who seek to employ a thermoregulation classification model on their species of interest should think carefully about the camera angles, colouration of studied animals (or marking colour if the animals are in captivity, like in the current framework) and backgrounds. Utilization of a more extensive training dataset, which includes multiple species, camera angles, illumination and backgrounds, can further improve accuracy rates at challenging illuminations and angles (Fennell et al., 2022). To adapt our framework to natural scenarios, we recommend first employing a general detection model on the collected images (e.g. Beery et al., 2019). Such models, trained on millions of images featuring various species in diverse scenarios, are highly effective at identifying animals in images, even under challenging conditions. After detection, species classification models can be utilized to determine the species of the detected animals (e.g. Schneider et al., 2020; Van Horn et al., 2018; Vecvanags et al., 2022). Finally, to classify the animal's microhabitat (sunny or shaded) and to understand its thermoregulation behaviour, one should train and apply a thermoregulation classification model as outlined in our framework.

With the rapid development of better and more robust computer vision approaches, thermoregulation classification will continue to improve and may include other forms of thermoregulation, such as increasing evaporative water loss (panting and gular flattering), increasing surface area (postural changes) and increasing insulation (e.g. feather fluffing or fur piloerection). However, these behaviours can be challenging to identify reliably due to poor camera settings, low image resolution or animal distance. Additionally, our model currently only distinguishes between two broad illumination conditions—sun and shade, and more nuanced lighting conditions may not be accurately classified (Palencia et al., 2022). Furthermore, the model's generalizability to other species and environments needs further exploration (Tabak et al., 2018). For example, longer animals, such as snakes, may position only part of their body in a warmer microhabitat (Regal, 1966). As with all image-based models, the accuracy will mostly depend on the image quality and the training dataset.

4.3 | Caveats and future directions

Although our model demonstrated promising performance, traditional approaches for tracking thermoregulation—such as manual observations, temperature loggers and telemetry—still offer significant benefits (e.g. Huey & Stevenson, 1979; Kerr et al., 2004; Moore et al., 2018). For instance, they allow for the continuous tracking of specific individuals' behaviours (Kerr et al., 2004). However, compared to our model, manual observations pose logistical challenges in studies covering extensive periods or multiple sites simultaneously (Weaver et al., 2021). Additionally, temperature loggers and telemetry involve capturing and sometimes anaesthetizing the animals, with the size of the animals limiting the feasibility of these methods and potentially the battery size of attached devices (Weaver et al., 2021; Wilson et al., 2021). Such studies may also exhibit bias towards individuals that are easier to capture (e.g. larger, slower, less vigilant; Michelangeli et al., 2016) and are unsuitable for rare species or early life stages. While using cameras addresses these issues, it introduces new challenges, such as the need for numerous cameras to capture a sufficient dataset and the difficulty in continuously tracking the same individual or differentiating between individuals. We propose that a combined approach could mitigate these method-specific challenges. For instance, camera-observed patterns of sun/shade usage could be validated with light-sensitive sensors on some population individuals. If these individuals show consistent patterns across both methods, it suggests that camera traps accurately capture the broader population's behaviour. Furthermore, for community studies, camera traps could focus on smaller species or those with smaller territories, while traditional methods could provide insights into larger or more mobile species. Analysing existing camera trap datasets can also highlight critical areas for conservation and habitat restoration efforts.

Understanding an animal's occupancy in sunny or shaded microhabitats is crucial but not sufficient for a comprehensive analysis of thermoregulation. Such analysis requires a deeper approach, comparing animals' microhabitat choices against available microclimates and contrasting observed body temperatures with those predicted by a null model for non-thermoregulating animals (Basson et al., 2017; Bauwens et al., 1996; Heath, 1964; Hertz et al., 1983; Huey & Pianka, 1977; Vickers & Schwarzkopf, 2016). High-resolution 3D mapping with drones facilitates accurate estimations of shade movement throughout the day, while remote sensing data identify shade-supplying elements like vegetation and rock cover (Stark et al., 2022), both essential for assessing microclimate impacts on thermoregulation. With microhabitats mapped, microclimate and biophysical models (Kearney & Porter, 2019) can then calculate the operative temperature for animals in various settings, a critical step in understanding their thermal challenges and thermoregulatory behaviour. Enriching this methodology, camera trap data analysis—for both captive and free-ranging animals—provides unique insights into microhabitat usage under varying conditions. This aids in designing climate-resilient enclosures that support efficient thermoregulation (de Azevedo et al., 2023; Wark et al., 2020) and in developing maps for microhabitats that promote activity under adverse climate conditions and should be prioritized in conservation or restoration programs.

4.4 | Summary

Our framework offers a compelling demonstration of the application of computer vision models in studying the thermal ecology of animals. As demonstrated in numerous camera-based studies, image collection can significantly alleviate the traditionally challenging task of manual observations. Yet, the vast volume of images generated by camera traps presents a new logistical challenge—sorting and understanding these images manually can be daunting and time-consuming. Given this challenge, our computer vision approach offers substantial benefits by automating the process of image analysis. With the urgency of understanding the thermal ecology of species in the face of climate change, the relative simplicity of using camera traps and the steady advancements in computer vision, our framework stands as a ripe opportunity for both ecologists aiming to comprehend microhabitat and microclimate preferences and physiologists seeking to gather essential data. Decades of camera-trap surveys worldwide have amassed a rich trove of images showcasing animals in diverse climates and habitats. We hope that our framework can assist in revealing the behavioural thermoregulation dimension hidden within these images and stimulate more computer-vision studies focused on microhabitat selection.

AUTHOR CONTRIBUTIONS

Ofir Levy conceived and designed the study and provided mentorship. Danny Mor and Ben Shermeister ran the experiment and collected data. Ben Shermeister also designed the study, collected the final data, trained the model and analysed the results. Ofir Levy and Ben Shermeister wrote the paper.

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CONFLICT OF INTEREST STATEMENT

There are no conflicts of interest among all authors for this manuscript.

DATA AVAILABILITY STATEMENT

All model output data, including all the data needed for creating the figures and tables, are available from Zenodo: https://doi.org/

10.5281/zenodo.11406382 (Levy & Shermeister, 2024). Updates to the codebase are available from https://github.com/levyofi/Shermeister_et_al_2024.

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